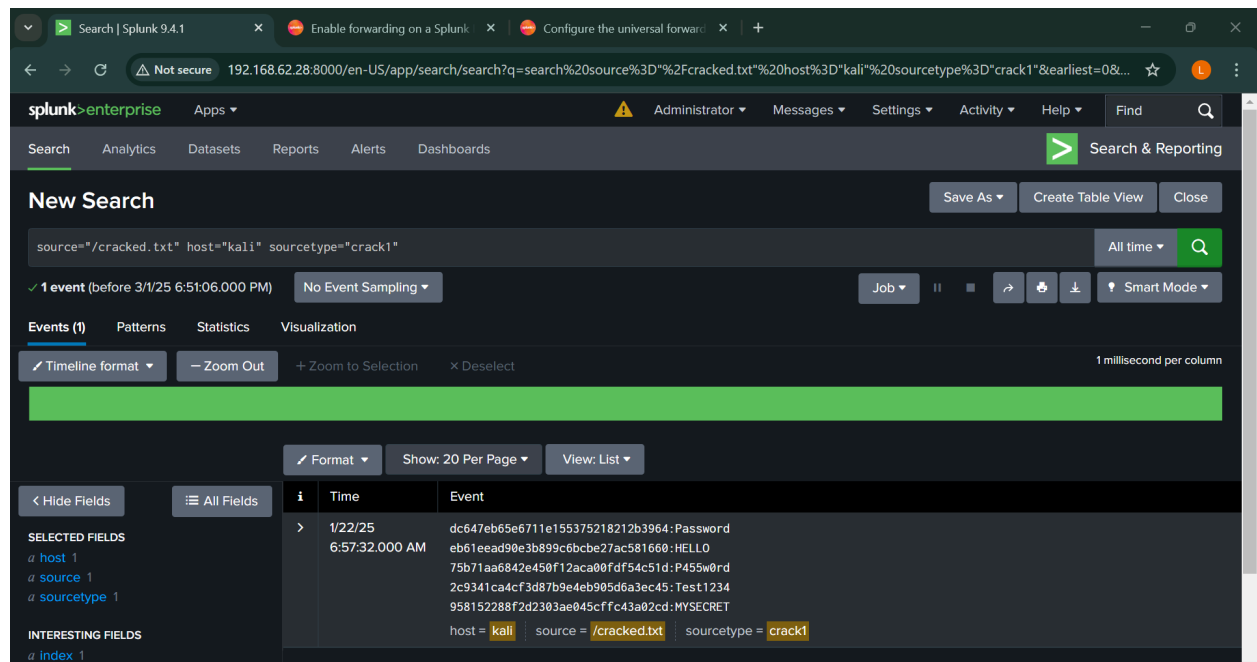




SECURITY INFORMATION AND EVENT MANAGEMENT (SIEM) USING SPLUNK.

"Centralized Log Collection and
Monitoring: Deploying Splunk
Universal Forwarder for Efficient
Data Ingestion and Using Splunk
Enterprise for Local Log Analysis
and Security Insights"

02.03.2025

Project Title

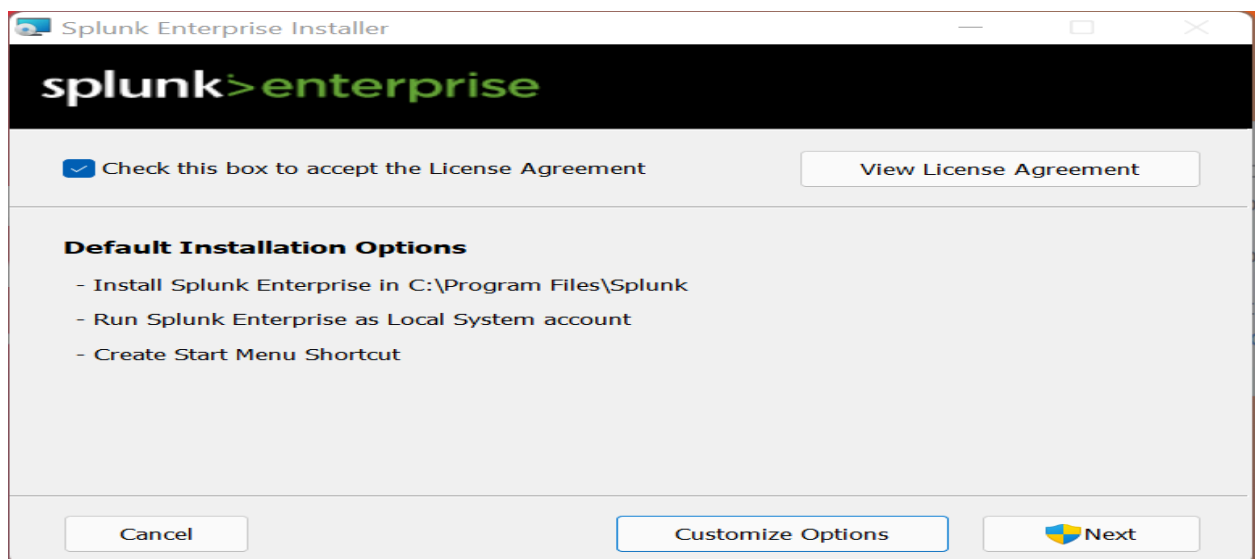
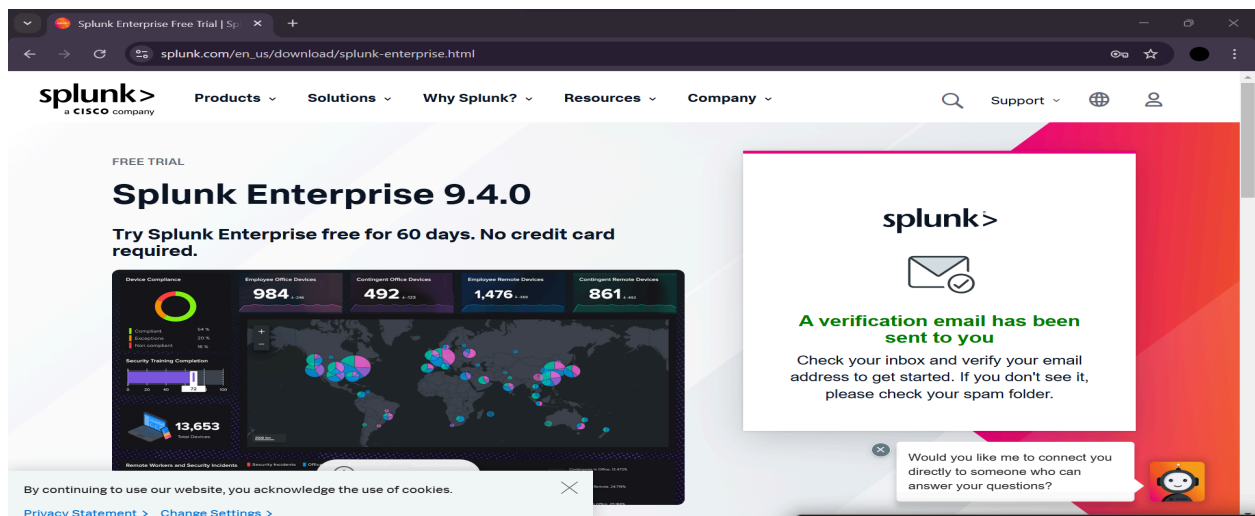
—Deploying Splunk Universal Forwarder for Efficient Data Ingestion and Utilizing Splunk Enterprise for Local Log Analysis and Security Insights

Executive Summary

In today's complex IT environments, organizations face the challenge of managing and analyzing vast amounts of log data generated by various systems and applications. Centralized log collection and monitoring are critical for ensuring operational efficiency, security, and compliance. This proposal outlines the deployment of Splunk Universal Forwarder for efficient data ingestion from multiple systems, combined with Splunk Enterprise for local log analysis and security insights. By leveraging these tools, we aim to provide a centralized, scalable, and secure solution for log management and real-time monitoring.

The project will focus on:

- 1. Installing and configuring Splunk Universal Forwarder** on Windows and Linux systems to collect and forward logs to a central Splunk server.
- 2. Deploying Splunk Enterprise** for local log analysis, enabling advanced search, reporting, and dashboard creation for security and operational insights.
- 3. Enhancing security monitoring** by integrating Windows event logs and other critical data sources into Splunk for real-time threat detection and incident response.



Project Objectives

The primary objectives of this project are:

Centralized Log Collection: Implement Splunk Universal Forwarder across Windows and Linux systems to collect and forward logs to a central Splunk server.

Efficient Data Ingestion: Ensure seamless and efficient ingestion of log data from multiple sources into Splunk Enterprise.

Local Log Analysis: Configure Splunk Enterprise to analyze local logs, providing actionable insights for operational and security teams.

Security Insights: Enable real-time monitoring of security events, including

Windows event logs, to detect and respond to potential threats.

Custom Dashboards: Create customizable dashboards in Splunk for visualizing key metrics and trends, facilitating quick decision-making.

Scope of Work

Deployment of Splunk Universal Forwarder

Download and Installation

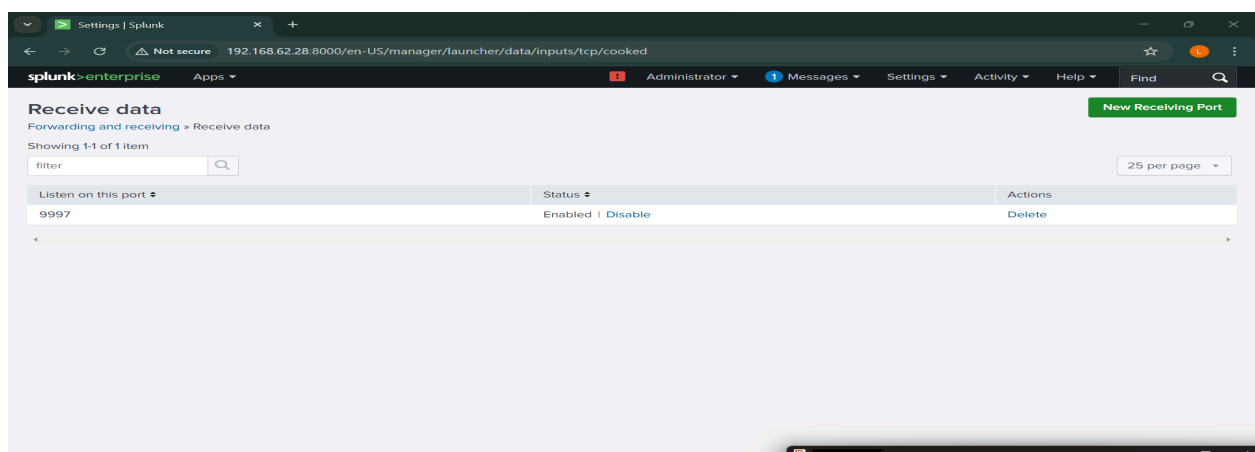
- Download the Splunk Universal Forwarder from the official Splunk website.
- Install the Universal Forwarder on Windows and Linux hosts, ensuring compatibility with the system architecture (e.g., 64-bit Windows).

-Configuration:

- Configure the Universal Forwarder to send logs to the central Splunk Enterprise server.
- Specify the IP address and port of the Splunk indexer (default port: 9997).
- Set up credentials for the Universal Forwarder to authenticate with the Splunk server.

-Log Collection:

- Configure the Universal Forwarder to collect specific logs, such as Windows event logs (Application, Security, System) and Linux system logs.
- Ensure logs are forwarded in real-time to the Splunk server for analysis.



Deployment of Splunk Enterprise

- Download and Installation:

- Download Splunk Enterprise from the official Splunk website.
- Install Splunk Enterprise on a local server (Windows or Linux).
- Create an admin account during installation for accessing the Splunk web interface.

-Configuration:

- Configure Splunk Enterprise to receive data from Universal Forwarders.
- Set up indexes for different log sources (e.g., Windows logs, Linux logs) to organize and manage data efficiently.
- Enable data inputs for local event logs (e.g., Application, Security, System logs on Windows).

- Log Analysis:

- Use Splunk's Search Processing Language (SPL) to query and analyze log data.
 - Create custom searches to monitor specific events, such as security breaches or system errors.
 - Generate reports and alerts based on log data to proactively address issues.
- Security Monitoring and Dashboards**

- Security Insights:

- Monitor critical security events, such as failed login attempts, audit log clears, and unauthorized access.
- Use Splunk's built-in security features to detect anomalies and potential threats.

- Custom Dashboards:

- Create dashboards to visualize key metrics, such as log volume, event types, and security incidents.
- Customize dashboards to display specific data relevant to different teams (e.g., IT, Security).
- Set up real-time alerts for critical events, ensuring timely response to security incidents.

splunk>enterprise 127.0.0.1:8000/en-US/app/search/search?q=search%20*&sid=1740518534.25&display.page.search.mode=smart&dispatch.sample_ratio=1&wor... Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

* Last 24 hours

✓ 5,082 events (2/24/25 10:00:00.000 PM to 2/25/25 10:22:14.000 PM) No Event Sampling Job

Events (5,082) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection × Deselect 1 hour per column

Format Show: 20 Per Page View: List

< Prev 1 2 3 4 5 6 7 8 ... Next >

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 3
- a sourcetype 3

INTERESTING FIELDS

- a Account_Domain 6
- a Account_Name 14

i	Time	Event
>	2/25/25 10:20:42.000 PM	02/25/2025 10:20:42 PM LogName=Security EventCode=4798 EventType=0 ComputerName= [redacted] Show all 27 lines host = [redacted] source = WinEventLog:Security sourcetype = WinEventLog:Security
>	2/25/25 10:20:42 PM	02/25/2025 10:20:42 PM

92°F Light rain 10:24 PM 2/25/2025

splunk>enterprise 127.0.0.1:8000/en-US/app/search/table?bs=%20host%3DLAO%20source%3D%20WinEventLog%3A%20Security%20EventCode%3D1102%20&dataset.se... Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

Create Table View

Cancel

* host [redacted] source="WinEventLog:Security" "EventCode=1102"

Previewing 2 events (10/31/24 7:27:11.000 PM to 2/25/25 10:50:20.000 PM) Sample: Latest

a source	a sourcetype	> _raw
WinEventLog:Security	WinEventLog:Security	> 02/25/2025 10:38:55 PM LogName=Security EventCode=1102 EventType=4 ComputerName=[redacted] SourceName=Microsoft-Windows-Eventlog Type=Information RecordNumber=324028 Keywords=Audit Success TaskCategory=Log clear OpCode=Info Message=The audit log was cleared. Subject: Security ID: S-1-5-21-1586029739-404 ...
WinEventLog:Security	WinEventLog:Security	> 02/25/2025 10:27:01 PM LogName=Security

Done

Snipping Tool

Technical Approach

Splunk Universal Forwarder Deployment

- Step 1: Download the Universal Forwarder from the Splunk website and install it on Windows and Linux hosts.
- Step 2: Configure the Universal Forwarder to send logs to the central Splunk server by specifying the IP address and port of the indexer.
- Step 3: Set up a log collection for specific sources, such as Windows event logs (Application, Security, System) and Linux system logs.
- Step 4: Verify that logs are being forwarded to the Splunk server in real-time.

Splunk Enterprise Deployment

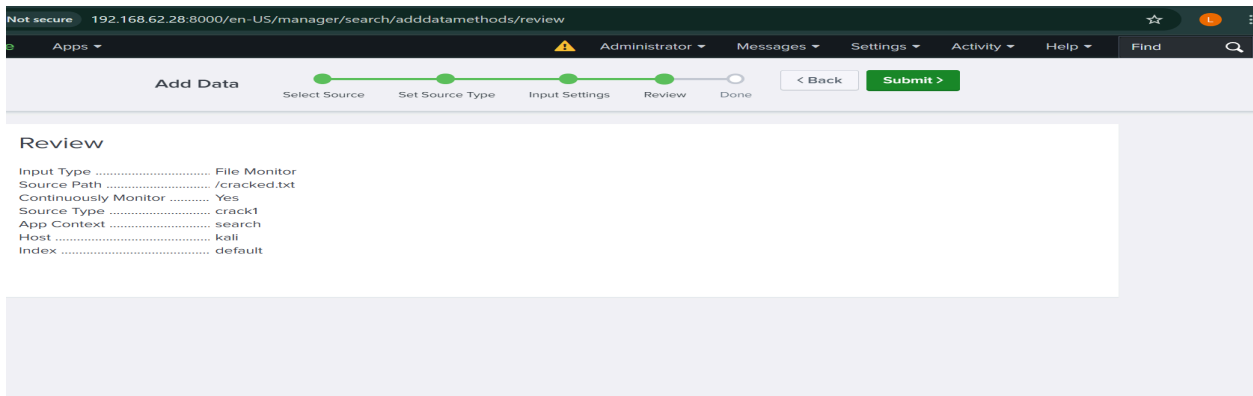
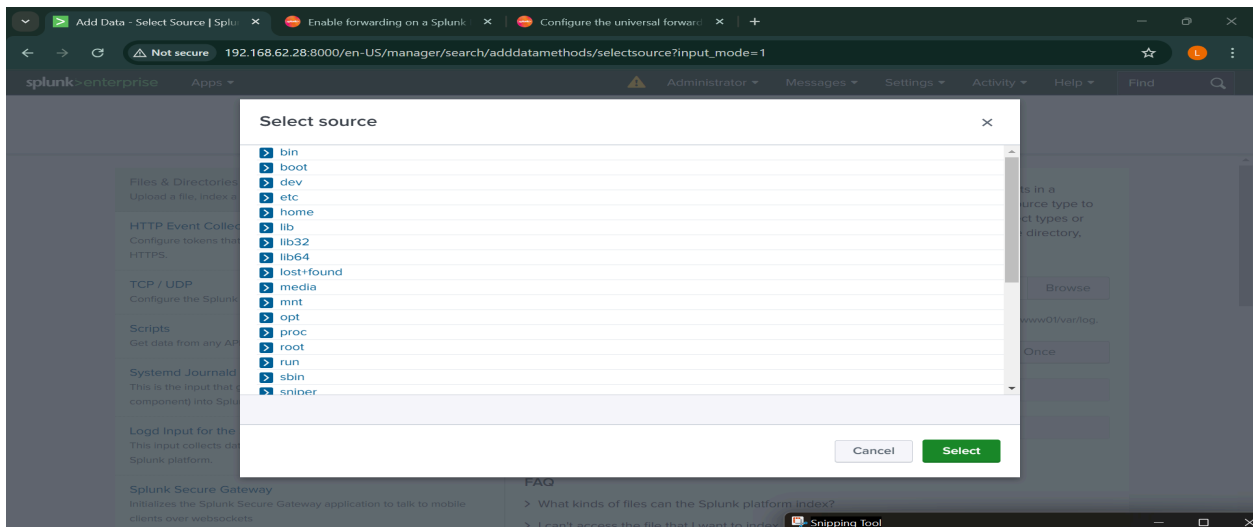
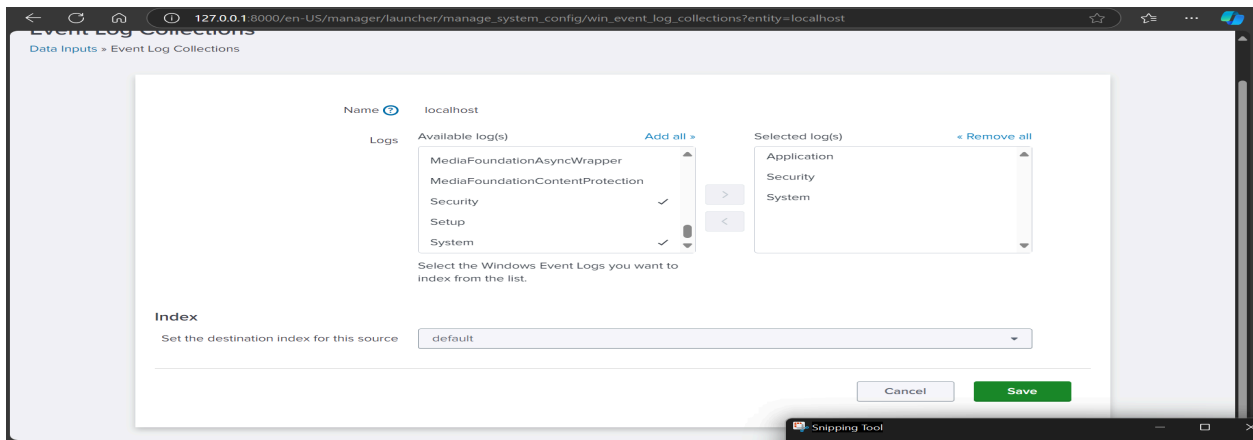
- Step 1: Download and install Splunk Enterprise on a local server.
- Step 2: Configure Splunk to receive data from Universal Forwarders by setting up data inputs and indexes.
- Step 3: Enable local event log collection for Windows and Linux systems.
- Step 4: Use SPL to create custom searches, reports, and alerts for monitoring and analysis.

Security Monitoring and Dashboards

- Step 1: Monitor critical security events using Splunk's built-in security features.
- Step 2: Create custom dashboards to visualize key metrics and trends.
- Step 3: Set up real-time alerts for critical events, ensuring timely response to security incidents.

Expected Outcomes

- Centralized Log Management: All logs from Windows and Linux systems will be collected and stored in a central Splunk server, providing a single source of truth for log data.
- Real-Time Monitoring: Real-time ingestion and analysis of logs will enable proactive monitoring of system and security events.
- Enhanced Security: By monitoring critical security events, the organization will be better equipped to detect and respond to potential threats.
- Customizable Dashboards: Custom dashboards will provide actionable insights for different teams, improving decision-making and operational efficiency.




```
kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)

Player
File Actions Edit View Help

# 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
Generating RSA private key, 2048 bit long modulus
.....+++++
e is 65537 (ex10001)
writing RSA key
Generating RSA private key, 2048 bit long modulus
.....+++++
e is 65537 (ex10001)
writing RSA key

Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.

[root@kali:~/home/kali]
# service splunk start

[root@kali:~/home/kali]
# service splunk status
splunk.service - LSB: Start splunk
Loaded: loaded (/etc/init.d/splunk; generated)
Active: active (running) since Sat 2025-03-01 03:53:22 EST; 11s ago
Invocation: 56392e2a52449856d729a13ba05d31
Docs: man:systemd-sysv-generator(8)
Process: 4457 ExecStart=/etc/init.d/splunk Start (code=exited, status=0/SUCCESS)
Tasks: 265 (limit: 1345)
Memory: 1.1G (peak: 1.2G)
CPU: 3min 59.982s
CGroup: /system.slice/splunk.service
└─589 [splunkd pid=588] splunkd -p 8089 start [process-runner]
    └─589 mongod --dbpath=/opt/splunk/var/lib/splunk/kvstore/mongo --storageEngine=wirefigner --wirefignerCacheSize=6000 --port=8191 --timestampFormat=iso8601-utc --oplogSize=200 --keyFile=/opt/splunk/var/lib/splunk/kvst
    └─589 /opt/splunk/bin/python3 -O /opt/splunk/lib/python3.9/site-packages/splunk/appserver/mrsparkle/root.py --proxied=127.0.0.1:8065,8080
    └─589 /opt/splunk/var/run/supervisor/pgk-run/pgk-identity251085010/identity
    └─589 [splunkd pid=588] [search-launcher] [process-runner]
    └─589 [splunkd pid=588] [search-launcher] [process-runner]

Mar 01 03:51:42 kali splunk[458]: All preliminary checks passed.
Mar 01 03:51:42 kali splunk[458]: Starting splunk server daemon (splunkd) ...
Mar 01 03:53:08 kali mongod[508]: looking for plugins in '/home/build/build-home/opt/mongo/lib/sasl2', failed to open directory, error: No such file or directory
Mar 01 03:53:08 kali mongod[508]: looking for plugins in '/home/build/build-home/opt/mongo/lib/sasl2', failed to open directory, error: No such file or directory
Mar 01 03:53:22 kali splunk[458]: waiting for web server at http://127.0.0.1:8000 to be available..... Done
Mar 01 03:53:22 kali splunk[458]: If you get stuck, we're here to help.
Mar 01 03:53:22 kali splunk[458]: Look for answers here: http://docs.splunk.com
Mar 01 03:53:22 kali splunk[458]: The Splunk web interface is at http://kali:18000
Mar 01 03:53:22 kali systemd[1]: Started splunk.service - LSB: Start splunk.

lines 1-31/31 (end)
```

Login | Splunk

192.168.62.28:8000/en-US/account/login?return_to=%2Fen-US%2F

splunk>enterprise

Username

Password

Sign In

First time signing in?

If you installed this instance, use the username and password you created at installation. Otherwise, use the username and password that your Splunk administrator gave you. If you've forgotten your username or password, please contact your Splunk administrator.

username admin

password The password you created when you installed this instance

First time signing in?

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phase	duration	deliverables
Planning and preparation	1 week	Project plan, resource allocation, and environment setup.
Splunk universal forwarder deployment	2 weeks	Universal Forwarder installed and configured on all target systems
Splunk enterprise deployment	2 weeks	Splunk Enterprise installed, configured, and integrated with Universal Forwarders.
Security Monitoring and Dashboards	1 week	Validate log collection, analysis, and dashboard functionality.
Testing and Validation	1 week	Validate log collection, analysis, and dashboard functionality.
Final Deployment and Handover	1 week	Final deployment, documentation, and handover to the operations team.

Conclusion

This project will provide the organization with a robust, centralized log collection and monitoring solution using Splunk Universal Forwarder and Splunk Enterprise. By implementing this solution, the organization will gain real-time visibility into system and security events, enabling proactive monitoring and rapid response to potential threats. The customizable dashboards and advanced search capabilities of Splunk will further enhance operational efficiency and decision-making.

Prepared by: LAO-TSE